

MUKHAMAD BHRUL ULUM

Automated Data Entry & Web Scraping Specialist | Python & AI Automation

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PROFESSIONAL SUMMARY

Results-driven Python Automation & Web Scraping Specialist with a strong analytical foundation in Accounting and Business Administration. I build production-grade automation pipelines that extract, clean, and deliver structured datasets at 10× the speed of manual work with zero human error - consistently processing 50,000+ records per project. From multi-target government tender scrapers to automated payroll engines and OCR document pipelines, I transform raw, unstructured data into clean, business-ready outputs that save clients hundreds of manual hours per month.

CORE COMPETENCIES

✓ Large-Scale Web Scraping & Data Mining ✓ Automated Data Entry & CRM Upload ✓ AI-Powered Data Cleaning & Validation
✓ Lead Generation & OSINT Pipelines ✓ OCR Document Processing ✓ Excel/PDF Automation ✓ LLM / AI Integration

TECHNICAL SKILLS

Web Scraping & Crawling	Selenium, BeautifulSoup, Scrapy, Requests, Anti-bot Bypass
Data Entry & Automation	Python Automation, Pandas, OpenPyXL, pyautogui, openpyxl
AI & LLM Integration	Prompt Engineering, LLM Pipelines, OCR (Tesseract), OpenCV
Backend & APIs	FastAPI, Python 3.12, Node.js, REST APIs, SQLite, JWT
Data Output & Storage	PostgreSQL, Redis, Excel/Google Sheets, PDF (ReportLab), MinIO
DevOps & Infrastructure	Docker, Nginx, Cloudflare, cPanel, HMAC-SHA256, QRIS Gateway

KEY SERVICES & EXPERTISE

► **Large-Scale Web Scraping & Data Mining:** Custom Python crawlers (Selenium + BeautifulSoup) targeting e-commerce, B2B directories, real estate, and JavaScript-heavy dynamic websites - delivering 10,000–100,000+ clean records per run.

► **Automated Data Entry & CRM Upload:** End-to-end pipelines that process, format, and push thousands of rows directly into Excel, Google Sheets, Airtable, or client CRMs - eliminating copy-paste entirely.

► **AI-Powered Data Cleaning & Validation:** Pandas + LLM integrations for deduplication, format standardization, messy-text repair, and 100% accuracy validation before delivery.

► **Lead Generation & OSINT Pipelines:** Extract and enrich B2B contact lists (emails, job titles, LinkedIn, company details) with multi-source cross-referencing for sales teams.

► **OCR & Document Data Extraction:** Convert scanned PDFs, images, and watermarked documents into clean structured databases using Tesseract OCR with custom pre-processing.

► **Excel & PDF Automation:** Generate professional multi-sheet Excel reports and formatted PDF documents (invoices, payslips, summaries) programmatically from raw data inputs.

PORTFOLIO PROJECTS

TalentScan — Automated HR Recruitment & Document Pipeline · Python · FastAPI · AI Integration · SMTP/IMAP · ReportLab

- Automated email ingestion pipeline that parses 200+ incoming applicant CVs directly from Gmail per batch, extracting structured candidate data (name, skills, experience) with zero manual input.
- AI scoring engine that evaluates CVs, interview responses, and quiz results to produce a unified competency score - reducing HR screening time by 85%.
- Auto-generated 50+ official PKWT employment contracts and formatted scoring reports per hiring cycle, eliminating all manual document preparation.
- Reduced full recruitment lifecycle from 5 days to under 6 hours for a 20-person hiring batch.

Tander — Government Tender Scraper & Intelligence Dashboard · Python · Selenium · Pandas · LLM · React · Node.js

- Built multi-target scraper covering 50+ LPSE government procurement portals across Indonesia, extracting 10,000+ tender records per weekly run into a clean central dashboard.
- Automated structured parsing of procurement documents — pulling project names, budgets, deadlines, and qualification requirements with 98%+ field accuracy.
- LLM-powered summarizer condenses 100-page tender documents into concise 1-page qualification checklists in under 30 seconds.
- Real-time multi-channel alert pipeline (Telegram, WhatsApp, Email) delivering deadline notifications and competitor bid-price intelligence to 30+ active users.

Hotel Database Scraper — Multi-Platform Travel Data Pipeline · Python · Selenium · BeautifulSoup · Airtable API · Groq/Gemini AI

- Built concurrent scrapers targeting Booking.com, Expedia, and Hotels.com - resolving dynamic hotel detail page URLs and extracting structured property data (name, rating, amenities, pricing, photos) at scale.
- Processed 5,000+ hotel records across 3 platforms into a unified Airtable database with AI-assisted field normalization (Groq/Gemini) for consistent data formatting.
- AI deduplication layer cross-referenced property names and geo-coordinates to eliminate 800+ duplicate entries across sources.
- Delivered a 100% clean, client-ready hotel database for a boutique travel media company - zero manual cleanup required.

Generator Dealer Scraper & OSINT Owner-Finding Pipeline · Python · Selenium · BeautifulSoup · DuckDuckGo/Bing API · Pandas

- Scraped 1,200+ generator dealer listings from multiple directories, extracting business names, addresses, contact numbers, and product specs into structured Excel output.
- Built an OSINT enrichment pipeline that cross-referenced dealer names against social media, business registries, and news sources to identify owner identities and direct contact details — achieving 70% owner match rate.
- Delivered a fully enriched B2B lead list ready for direct sales outreach, reducing manual research time from 3 weeks to 48 hours.

Faire Bulk-Upload Automation — E-Commerce Data Entry Pipeline · Python · Streamlit · Pandas · Selenium · OpenPyXL

- Automated bulk product upload workflow for a dress manufacturer client on the Faire B2B platform - processing 500+ SKUs per session from raw Excel input with zero manual copy-pasting.
- Streamlit dashboard allowed non-technical client staff to trigger uploads, monitor progress, and review error logs independently.
- Reduced product listing time from 3 days of manual entry to under 2 hours per catalog update, with built-in data validation catching format errors before submission.

RBV CLEAN — Advanced OCR & PDF Data Extraction Pipeline · Python · CustomTkinter · Tesseract OCR · OpenCV · ocrmypdf

- Engineered a desktop OCR application processing 200+ scanned university module PDFs - including automated watermark detection and removal using LAB color space thresholding.
- Custom image pre-processing pipeline (morphological operations, deskewing, contrast normalization) boosted OCR character accuracy from 61% to 94% on low-quality scans.
- Output: clean, fully searchable text databases from previously unstructured scanned documents - ready for downstream indexing or data entry workflows.

Invoice & Payroll Automation System — Python Desktop App · Python · Tkinter · ReportLab · OpenPyXL · SQLite

- Built a full desktop automation system handling end-to-end invoice lifecycle (Draft → Sent → Paid) and Indonesian payroll calculation (BPJS, PPh 21 progressive, overtime) for 50+ employees.
- Auto-generates professional PDF invoices and payslips via ReportLab, and exports multi-sheet Excel payroll reports via OpenPyXL - eliminating 40+ hours/month of manual HR admin work.
- Compliant with Indonesian 2024 tax regulations: PTKP brackets, TER method PPh 21, BPJS Kesehatan & Ketenagakerjaan deductions calculated automatically.

PROFESSIONAL EXPERIENCE

Independent Python Automation & Data Specialist

Self-Employed / Freelance | 2024 – Present

- Design and deploy end-to-end Python automation and web scraping pipelines for clients across data entry, lead generation, HR automation, and document processing.
- Deliver production-grade scrapers processing 10,000 - 100,000+ records per run with structured output to Excel, Airtable, PostgreSQL, or client CRMs.
- Integrate LLM/AI APIs to automate data cleaning, document summarization, and intelligent field extraction from unstructured sources.
- Leverage Accounting/Business Administration background to ensure all data outputs meet strict accuracy, auditability, and business-logic standards.

EDUCATION

Bachelor of Business Administration

Universitas Terbuka Indonesia | 2026 – Present

Academic Background in Accounting

Focus: systematic ledgers, financial auditing compliance, structural data classification - directly applicable to data quality, validation, and business-logic automation.

LANGUAGES

Indonesian: Native | **English:** Professional Working Proficiency

AVAILABILITY & REMOTE WORK

Available immediately for remote freelance or contract roles. Experienced working asynchronously across time zones with clear documentation and handover. All deliverables include clean, production-ready code with full documentation.